

excluded from the analysis, therefore the results are not related to data manipulation or "data fiddling" (Scargle, 2002).

Related factors. Though the results are supportive of our exploratory predictions, at this point, it cannot be definitively concluded that the results are due to an influence of group meditation and/or the Vedic Observatory. Other factors besides meditation or related auxiliary factors to meditation could be involved. Further research could include ruling out the simple effect of large numbers of people in silence, or numbers of people sitting non-actively. However, no significance has been found for relatively silent non-mobile audiences at conferences (Nelson et al., 1998).

Lag effect. The post-test results of this study could be interpreted as a candidate for a carryover residual effect, lag effect or entrainment effect. Could using the RNG, during the meditations or at the Vedic Observatory create a lag effect or alter the results of RNG? A new RNG (Orion) developed possible cumulative drifts after exposure to the group meditation, but not before. Intentional time delay effects have been previously reported in the literature involving single subject studies (Dunne and Jahn, 1992). A time lag or carryover effect that diminishes over months has also been measured in studies evaluating the effect of group meditation on societal indexes (Dillbeck, 1990; Dillbeck et al., 1987; Hagelin et al., 1999; Orme-Johnson et al., 1988). Extensive longitudinal research would be needed to support or dismiss the lag effect or entrainment effect as an alternative explanation of the results.

Co-author Radin, reports continued randomness after up to 5 years of use with multiple Orion RNGs when run non-xored. However two of these RNGs were used for experiments involving meditation. Co-author Radin notes these two RNGs then developed in the post-test a downward drift similar to that reported in the present experiment. Preliminary reviews found no downward drift in subsequent non-meditation related experiments with these RNGs. Radin's investigation and reanalysis of this previous data (Radin 2006; Radin & Atwater, 2006) is underway in order to discover if there is a meditation and RNG interaction responsible for the drift in the mean/variance or a lag effect or some other possible mundane answer.

It is not clear there is a cumulative drift involved or what the source of the drift is or if there is a lag effect. At this point, the difference in post-test data from pre-test baseline "control" data is not clearly accounted for, therefore a statistical control for any cumulative drift regardless of the source was used. The experimental data was still significant after reanalysis for a possible drift for the meditation, yogic flying and Vedic Observatory for both the initial experiment and its replication.

Conclusion

Our predictions for the meditation data, yogic flying and Vedic observatory data were significantly supported and were in the predicted direction. Our work adds to the premise that certain activities that foster transcendental experiences

(Alexander & Langer, 1990; Mason et al, 1997; Orme-Johnson et al., 1988; Travis et al, 2002) may reflect a more decreasing directional trend (increased proportion of zeros) in RNG outputs. Alternative explanations do not clearly account for the observed results. The results were still significant even after controlling for a possible cumulative drift of the mean from an unknown source.

To our knowledge this is the first experiment with specific predictions for the direction of a mean shift, and it involves the largest number of synchronized meditations recorded with a local RNG on site. Having a population doing a standardized mental technique on a regular basis is advantageous in studying various aspects of the phenomenon. Further research appears warranted to explore group meditation as a venue for anomalous results with the RNG. Future research could test the direction of the results, distance effects from the group, possible lag or entrainment effects, experimenter effect, non-xoring data techniques, group size effects, number of RNGs and possible auxiliary factors. Theoretical questions could include a continued inquiry (Hagelin, 1987; Nader, 2000; Nelson, 2002d; Radin, 2002; Routt, 2005) as to whether or not consciousness is a causal factor.

What is the possible practical contributions and application of this research? It is conceivable that RNGs could be used to indicate directional changes in a proposed global collective consciousness. Just as changes in seismic meters are used to detect high and low indications of impending earthquakes, RNG outputs could warn us of changes in collective consciousness while considering any anticipatory effects. RNGs could also be employed to evaluate preventive and ameliorative measures that utilize collective consciousness. For example, the RNG could evaluate the efficacy of various technologies from many traditions, including group meditations to reduce collective stress in global consciousness in order to prevent and reduce local and global tragedies.

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